

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics
Practical File – ODD Semester 2026-27

StudentID: 24DIT066
Student Name: Sil Shah

Practical 6

1. Problem Definition

Scenario:

ShopSmart Analytics Pvt. Ltd. wants to analyze its e-commerce sales data. The company needs to identify high-performing product categories, top-selling products and revenue contribution by city. Since the amount of data is large, Apache Spark is used for scalable data processing and analysis.

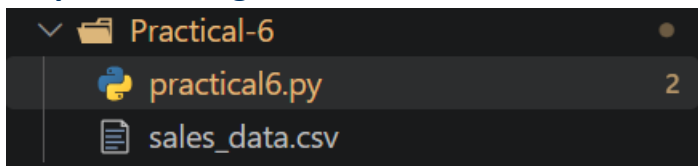
Objective

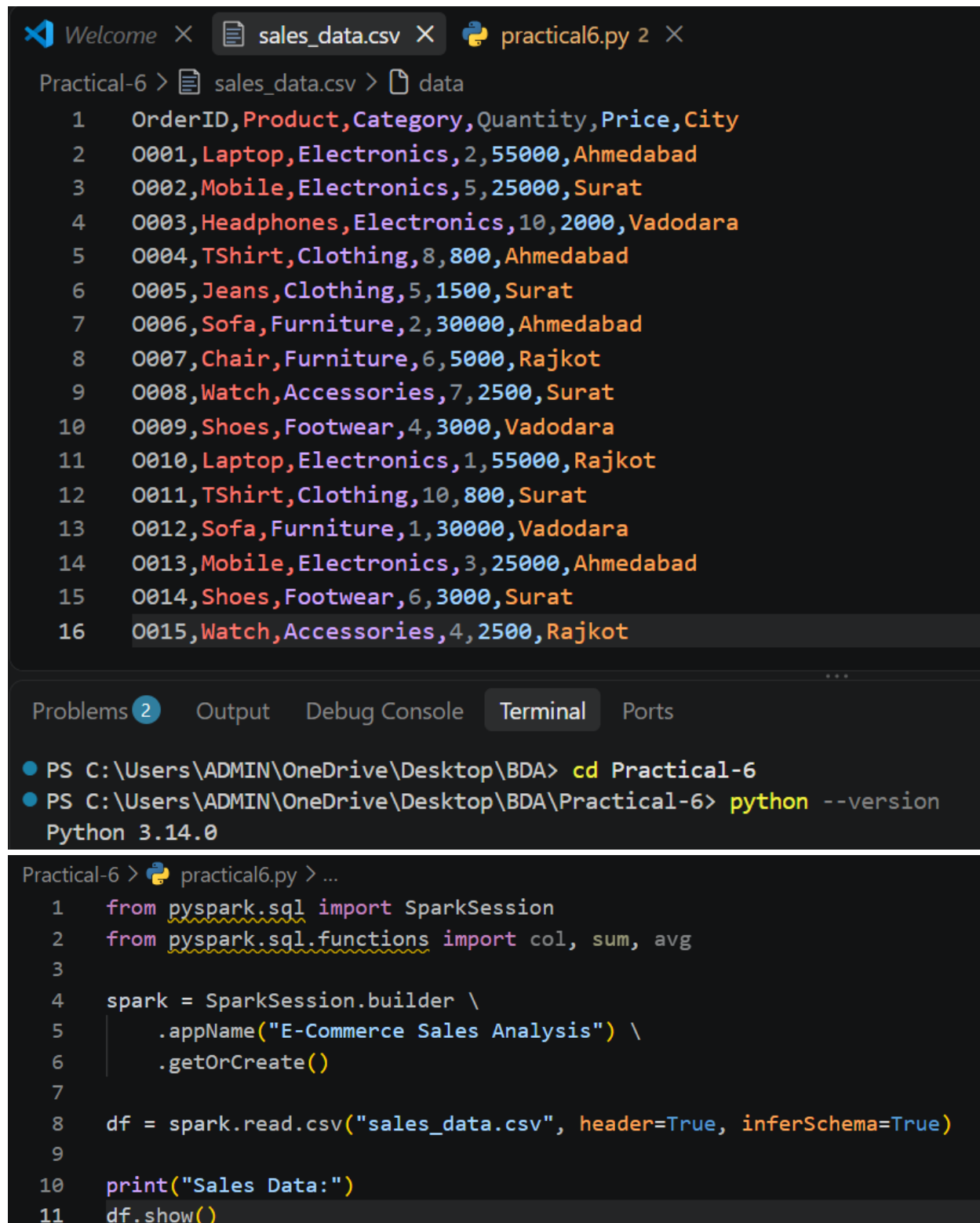
To use PySpark DataFrames to:

- Load e-commerce sales data
- Validate the dataset
- Calculate revenue
- Find category-wise revenue
- Find top-selling products
- Calculate city-wise revenue
- Sort categories according to revenue
- Generate analytical summaries
- Extract business insights

PART A — Practical Execution

Step 1: Creating Folder and files





The image shows a VS Code editor interface with two tabs: 'sales_data.csv' and 'practical6.py 2'. The 'sales_data.csv' tab is active, displaying a CSV file with 16 rows of data. The data includes columns for OrderID, Product, Category, Quantity, Price, and City. The 'practical6.py' tab is also visible, showing a Python script that uses PySpark to read the CSV file and display its contents.

```
Practical-6 > sales_data.csv > data
```

OrderID	Product	Category	Quantity	Price	City
0001	Laptop	Electronics	2	55000	Ahmedabad
0002	Mobile	Electronics	5	25000	Surat
0003	Headphones	Electronics	10	2000	Vadodara
0004	TShirt	Clothing	8	800	Ahmedabad
0005	Jeans	Clothing	5	1500	Surat
0006	Sofa	Furniture	2	30000	Ahmedabad
0007	Chair	Furniture	6	5000	Rajkot
0008	Watch	Accessories	7	2500	Surat
0009	Shoes	Footwear	4	3000	Vadodara
0010	Laptop	Electronics	1	55000	Rajkot
0011	TShirt	Clothing	10	800	Surat
0012	Sofa	Furniture	1	30000	Vadodara
0013	Mobile	Electronics	3	25000	Ahmedabad
0014	Shoes	Footwear	6	3000	Surat
0015	Watch	Accessories	4	2500	Rajkot

```
Problems 2 Output Debug Console Terminal Ports
```

- PS C:\Users\ADMIN\OneDrive\Desktop\BDA> cd Practical-6
- PS C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-6> python --version

Python 3.14.0

```
Practical-6 > practical6.py > ...
```

```
1 from pyspark.sql import SparkSession
2 from pyspark.sql.functions import col, sum, avg
3
4 spark = SparkSession.builder \
5     .appName("E-Commerce Sales Analysis") \
6     .getOrCreate()
7
8 df = spark.read.csv("sales_data.csv", header=True, inferSchema=True)
9
10 print("Sales Data:")
11 df.show()
```

2. Output of Practical6.py

===== ORIGINAL DATA =====

OrderID	Product	Category	Quantity	Price	City
0001	Laptop	Electronics	2	55000	Ahmedabad
0002	Mobile	Electronics	5	25000	Surat
0003	Headphones	Electronics	10	2000	Vadodara
0004	TShirt	Clothing	8	800	Ahmedabad
0005	Jeans	Clothing	5	1500	Surat
0006	Sofa	Furniture	2	30000	Ahmedabad
0007	Chair	Furniture	6	5000	Rajkot
0008	Watch	Accessories	7	2500	Surat
0009	Shoes	Footwear	4	3000	Vadodara
0010	Laptop	Electronics	1	55000	Rajkot
0011	TShirt	Clothing	10	800	Surat
0012	Sofa	Furniture	1	30000	Vadodara
0013	Mobile	Electronics	3	25000	Ahmedabad
0014	Shoes	Footwear	6	3000	Surat
0015	Watch	Accessories	4	2500	Rajkot

===== SCHEMA =====

root

```
-- OrderID: string (nullable = true)
-- Product: string (nullable = true)
-- Category: string (nullable = true)
-- Quantity: integer (nullable = true)
-- Price: integer (nullable = true)
-- City: string (nullable = true)
```

===== DATA TYPES =====

Column: OrderID	Data Type: string
Column: Product	Data Type: string
Column: Category	Data Type: string
Column: Quantity	Data Type: int
Column: Price	Data Type: int
Column: City	Data Type: string

```
===== REVENUE CALCULATION =====
```

OrderID	Product	Category	Quantity	Price	City	Revenue
0001	Laptop	Electronics	2	55000	Ahmedabad	110000
0002	Mobile	Electronics	5	25000	Surat	125000
0003	Headphones	Electronics	10	2000	Vadodara	20000
0004	TShirt	Clothing	8	800	Ahmedabad	6400
0005	Jeans	Clothing	5	1500	Surat	7500
0006	Sofa	Furniture	2	30000	Ahmedabad	60000
0007	Chair	Furniture	6	5000	Rajkot	30000
0008	Watch	Accessories	7	2500	Surat	17500
0009	Shoes	Footwear	4	3000	Vadodara	12000
0010	Laptop	Electronics	1	55000	Rajkot	55000
0011	TShirt	Clothing	10	800	Surat	8000
0012	Sofa	Furniture	1	30000	Vadodara	30000
0013	Mobile	Electronics	3	25000	Ahmedabad	75000
0014	Shoes	Footwear	6	3000	Surat	18000
0015	Watch	Accessories	4	2500	Rajkot	10000

```
===== CATEGORY-WISE REVENUE =====
```

Category	Total_Revenue
Electronics	385000
Furniture	120000
Footwear	30000
Accessories	27500
Clothing	21900

```
===== TOP-SELLING PRODUCTS =====
```

Product	Total_Quantity_Sold
TShirt	18
Watch	11
Shoes	10
Headphones	10
Mobile	8
Chair	6
Jeans	5
Laptop	3
Sofa	3

===== DISCOUNT ANALYSIS =====

OrderID	Product	Category	Quantity	Price	City	Revenue	Discount	Final_Revenue
0001	Laptop	Electronics	2	55000	Ahmedabad	110000	11000.0	99000.0
0002	Mobile	Electronics	5	25000	Surat	125000	12500.0	112500.0
0003	Headphones	Electronics	10	2000	Vadodara	20000	2000.0	18000.0
0004	TShirt	Clothing	8	800	Ahmedabad	6400	640.0	5760.0
0005	Jeans	Clothing	5	1500	Surat	7500	750.0	6750.0
0006	Sofa	Furniture	2	30000	Ahmedabad	60000	6000.0	54000.0
0007	Chair	Furniture	6	5000	Rajkot	30000	3000.0	27000.0
0008	Watch	Accessories	7	2500	Surat	17500	1750.0	15750.0
0009	Shoes	Footwear	4	3000	Vadodara	12000	1200.0	10800.0
0010	Laptop	Electronics	1	55000	Rajkot	55000	5500.0	49500.0
0011	TShirt	Clothing	10	800	Surat	8000	800.0	7200.0
0012	Sofa	Furniture	1	30000	Vadodara	30000	3000.0	27000.0
0013	Mobile	Electronics	3	25000	Ahmedabad	75000	7500.0	67500.0
0014	Shoes	Footwear	6	3000	Surat	18000	1800.0	16200.0
0015	Watch	Accessories	4	2500	Rajkot	10000	1000.0	9000.0

===== LEAST-PERFORMING CATEGORY =====

Category	Total_Final_Revenue
Clothing	19710.0

only showing top 1 row

===== MONTHLY REVENUE =====

Monthly analysis requires an OrderDate column in the dataset. The dataset does not contain any date column.

PS C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-6> ERROR: The process with PID 18096 (child process of PID 21872) could not be terminated.

Reason: There is no running instance of the task.

SUCCESS: The process with PID 21872 (child process of PID 2852) has been terminated.

SUCCESS: The process with PID 2852 (child process of PID 25932) has been terminated.

3. Create Revenue Column

```
===== REVENUE CALCULATION =====
```

OrderID	Product	Category	Quantity	Price	City	Revenue
0001	Laptop	Electronics	2	55000	Ahmedabad	110000
0002	Mobile	Electronics	5	25000	Surat	125000
0003	Headphones	Electronics	10	2000	Vadodara	20000
0004	TShirt	Clothing	8	800	Ahmedabad	6400
0005	Jeans	Clothing	5	1500	Surat	7500
0006	Sofa	Furniture	2	30000	Ahmedabad	60000
0007	Chair	Furniture	6	5000	Rajkot	30000
0008	Watch	Accessories	7	2500	Surat	17500
0009	Shoes	Footwear	4	3000	Vadodara	12000
0010	Laptop	Electronics	1	55000	Rajkot	55000
0011	TShirt	Clothing	10	800	Surat	8000
0012	Sofa	Furniture	1	30000	Vadodara	30000
0013	Mobile	Electronics	3	25000	Ahmedabad	75000
0014	Shoes	Footwear	6	3000	Surat	18000
0015	Watch	Accessories	4	2500	Rajkot	10000

4. Calculate Category-wise Revenue

```
===== CATEGORY-WISE REVENUE =====
```

Category	Total_Revenue
Electronics	385000
Furniture	120000
Footwear	30000
Accessories	27500
Clothing	21900

5. Find Top-Selling Products

```
===== TOP-SELLING PRODUCTS =====
+-----+-----+
| Product|Total_Quantity_Sold|
+-----+-----+
| TShirt|          18|
| Watch|          11|
| Shoes|          10|
| Headphones|        10|
| Mobile|           8|
| Chair|           6|
| Jeans|           5|
| Laptop|           3|
| Sofa|           3|
+-----+-----+
```

6. Calculate City-wise Revenue

```
===== CITY-WISE REVENUE =====
+-----+-----+
| City|Total_Revenue|
+-----+-----+
| Ahmedabad|    251400|
| Surat|    176000|
| Rajkot|    95000|
| Vadodara|    62000|
+-----+-----+
```

7. Top 3 Revenue-Generating Cities and Average Revenue per Order

```
===== TOP 3 CITIES =====
+-----+-----+
| City|Total_Revenue|
+-----+-----+
| Ahmedabad|    251400|
| Surat|    176000|
| Rajkot|    95000|
+-----+-----+
only showing top 3 rows
===== AVERAGE REVENUE PER ORDER =====
+-----+-----+
| Average_Revenue_Per_Order|
+-----+-----+
|          38960.0|
+-----+-----+
```

8. Add Discount Column

```
===== DISCOUNT ANALYSIS =====
```

OrderID	Product	Category	Quantity	Price	City	Revenue	Discount	Final_Revenue
0001	Laptop	Electronics	2	55000	Ahmedabad	110000	11000.0	99000.0
0002	Mobile	Electronics	5	25000	Surat	125000	12500.0	112500.0
0003	Headphones	Electronics	10	2000	Vadodara	20000	2000.0	18000.0
0004	TShirt	Clothing	8	800	Ahmedabad	6400	640.0	5760.0
0005	Jeans	Clothing	5	1500	Surat	7500	750.0	6750.0
0006	Sofa	Furniture	2	30000	Ahmedabad	60000	6000.0	54000.0
0007	Chair	Furniture	6	5000	Rajkot	30000	3000.0	27000.0
0008	Watch	Accessories	7	2500	Surat	17500	1750.0	15750.0
0009	Shoes	Footwear	4	3000	Vadodara	12000	1200.0	10800.0
0010	Laptop	Electronics	1	55000	Rajkot	55000	5500.0	49500.0
0011	TShirt	Clothing	10	800	Surat	8000	800.0	7200.0
0012	Sofa	Furniture	1	30000	Vadodara	30000	3000.0	27000.0
0013	Mobile	Electronics	3	25000	Ahmedabad	75000	7500.0	67500.0
0014	Shoes	Footwear	6	3000	Surat	18000	1800.0	16200.0
0015	Watch	Accessories	4	2500	Rajkot	10000	1000.0	9000.0

9. Least-Performing Category

```
===== LEAST-PERFORMING CATEGORY =====
```

Category	Total_Final_Revenue
Clothing	19710.0

only showing top 1 row

10. Monthly Revenue Summary

Our current dataset does not contain a date/month column.

So for this supplementary question, we would need to extend the dataset with an OrderDate column.

11. Key Questions / Analysis

Q1. Why are Spark DataFrames preferred over traditional spreadsheets for large-scale analytics?

Spark DataFrames are preferred because they can process very large datasets efficiently using distributed computing. Data can be divided into partitions and processed in parallel across multiple machines. Spreadsheets are more suitable for small or moderate-sized datasets and have limitations when the data becomes very large.

Q2. What is the role of the groupBy() operation in data aggregation?

The `groupBy()` operation groups records having the same value in a particular column. It is commonly used with aggregation functions such as `sum()`, `avg()`, `count()`, `min()` and `max()`.

For example:

```
df.groupBy("Category").sum("Revenue")
```

groups the sales according to product category and calculates the total revenue for each category.

Q3. Explain the difference between transformations and actions in Spark.

Transformations create a new DataFrame or RDD from existing data. They are lazily evaluated.

Examples:

`select()`

`filter()`

`withColumn()`

`groupBy()`

Actions trigger the actual execution of Spark operations and return a result or output.

Examples:

`show()`

`count()`

`collect()`

Therefore:

Transformation → Defines the operation

Action → Executes the operation

Q4. How does Spark improve scalability for e-commerce analytics?

Spark improves scalability by using distributed and parallel processing. Large e-commerce datasets can be divided into partitions and processed simultaneously across multiple computing resources. Spark DataFrames also provide optimized execution, making large-scale sales analysis faster and more efficient.

Q5. Which category contributes the highest revenue and what business action can be taken?

The highest-revenue category is identified from the **category-wise revenue analysis** by sorting categories in descending order of revenue.

Highest Revenue Category: _____

A suitable business action is to maintain sufficient inventory for high-performing products, continue targeted marketing campaigns, and analyze customer demand to maintain the category's performance.

12. Conclusion

The practical successfully demonstrated the use of **Apache Spark DataFrames for e-commerce sales analysis**. The dataset was loaded and validated, and important business metrics such as revenue, category-wise sales, product quantity, and city-wise revenue were calculated.

The use of operations such as `withColumn()`, `groupBy()`, `sum()`, `avg()`, `orderBy()` and `limit()` helped perform efficient data analysis. Spark provides a scalable approach for processing large e-commerce datasets and generating useful business insights.

The analysis can help businesses identify high-performing categories and products, understand geographical revenue distribution, improve marketing strategies, and make data-driven decisions.

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13. Github Link

Github Link: [24DIT066-tech/Practical-6 at main · sil-sys/24DIT066-tech](https://github.com/24DIT066-tech/Practical-6)